

# Fauzan Mohammed

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## SUMMARY

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Experienced Data Scientist and NLP Engineer with a proven track record in developing production multi-agent LLM systems and responsible AI. Skilled in designing and deploying NLP solutions that enhance customer experiences and drive business value. Adept at bridging technical expertise with real-world applications to solve high-value **customer problems**.

## EDUCATION

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**Rensselaer Polytechnic Institute**

*Masters in IT/CS*

Troy, NY

*Aug 2023 – Dec 2024*

**Kwame Nkrumah University of Science and Technology**

*Bachelor of Science in Mechanical Engineering*

Kumasi, Ghana

*Sep 2016 – June 2020*

## EXPERIENCE

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**Data Scientist – NLP/LLM Systems + Responsible AI**

*Walmart Global Tech*

Aug 2024 – Present

*Sunnyvale, CA*

- Contributed to the design and deployment of a production multi-agent LLM system powering Walmart's conversational shopping platform, orchestrating intelligent search, guided discovery, and personalized recommendations for 20M+ monthly active users.
- Designed prompt-tuned LLM-as-judge evaluation frameworks for search quality and product relevance assessment across raw and reranked results, driving measurable improvements in ranking precision across product categories.
- Built production guardrail infrastructure using prompt-tuned LLM judges on GCP and Element Gateway, implementing real-time safety enforcement and hallucination detection to ensure responsible AI compliance for 10M+ users.
- Designed and maintained prompt templates, multi-step chains, and persona configurations across agent workflows, implementing version-controlled prompt libraries for consistency and reuse across the conversational platform.
- Developed an AI agent-based Event Title Generator leveraging LLM function calling to trigger internal APIs, autonomously infer user intent, map to structured event categories, and generate personalized titles within the shopping experience.
- Architected multi-turn conversation management logic including memory handling, contextual continuity, and dynamic persona assignment across domain-specific agent roles.
- Developed unsupervised NLP pipeline using DBSCAN clustering for topic modeling and intent extraction, expanding chatbot query coverage from 15% to 74%.

**Data Scientist, Intern**

*Walmart Global Tech*

May 2024 – Aug 2024

*Sunnyvale, CA*

- Engineered Transformer-based fallback detection system using semantic clustering to classify 100K+ daily fallback queries into 27 intent clusters, improving diagnostic accuracy by 6%.
- Optimized LLM router architecture through prompt engineering and query decomposition, improving task-selection accuracy by 4% and reducing latency by 18%.
- Built automated evaluation dashboards using SQL and Python for real-time monitoring of precision, recall, and prompt drift, reducing experimentation cycles by 30%.

**Junior Machine Learning Engineer**

*Omdena Thrive*

Dec 2022 – Apr 2023

*Remote*

- Engineered BERT-based NLP models with attention mechanisms achieving 85% classification accuracy on

5K+ mental health journal entries; deployed production analytics dashboard on GCP using Flask and Plotly for 200+ clinical users.

PROJECTS

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**AI-Powered Financial Rule Automation System**

Dec 2024

- Built AI rule-generation engine integrating multiple Transformer models (MiniLM, RoBERTa, DeBERTa) with QuickBooks API, implementing multi-task fusion architecture to auto-categorize transactions at 92% precision and reduce manual rule creation by 70%.

LEADERSHIP

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**Co-founder and Director**

*Fama Foundation*

Aug 2019 – Present

*Accra, Ghana*

- Led impactful programs across 10 Ghanaian communities, enrolling 325 teenage girls in computer programming and literacy education.

TECHNICAL SKILLS

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**Languages:** Python, SQL, Java

**ML/DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, spaCy, NLTK

**LLM & AI:** LangChain, LlamaIndex, FAISS, OpenAI/Anthropic/Gemini APIs

**Cloud & Data:** GCP (Vertex AI, BigQuery, Cloud Run), Docker, Kubernetes, Pandas, NumPy, Spark

**MLOps:** MLflow, Weights & Biases, Flask, FastAPI, CI/CD, Git, DVC

**Specialties:** Multi-Agent Systems, LLM Fine-Tuning (SFT, RLHF, DPO), Prompt Engineering, RAG, Model Evaluation, Responsible AI, Function Calling & Tool Use, Conversation Flow Design